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Operational AI Embedding and Green Innovation: When Rapid Adaptation Matters --Manuscript Draft--

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Operational AI Embedding and Green Innovation: When Rapid Adaptation Matters

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ABSTRACT

Firms increasingly report using artificial intelligence (AI), but disclosure alone does not show whether AI is embedded in the workflows where operating decisions are monitored, coordinated, and revised. We introduce disclosed AI-enabled execution capability (AEC), a sentence-level construct that treats annual-report text as evidence of workflow embedding only when AI language is tied to concrete execution features and operational or governance processes. Using 19,018 firm-year observations for Chinese A-share listed firms from 2020 to 2025, we examine AEC and subsequent green invention patenting. After firm fixed effects, prior innovation, and adjacent digital measures are introduced, the average within-firm AEC association is small. In the benchmark-adjusted model, the AEC $\times$ real-time adaptation requirement interaction is 0.066 (SE = 0.028; Benjamini–Hochberg $q = 0.074$), while the corresponding coordination interaction is 0.051. The equality test gives $p = 0.091$. The same directional pattern appears under timing restrictions, non-text task proxies, and an external measure of environmental-enforcement pressure. The evidence is strongest when operating conditions require rapid adaptation. The design is observational. Disclosed AEC is measured from corporate reports and does not verify system use.

1. Introduction

Recent firm-level studies link AI adoption to corporate green innovation in Chinese listed firms (Hussain et al., 2024). Text-based measures of AI applications report a similar association (Xi and Shao, 2025), while policy-induced AI adoption has also been associated with increases in the quantity and quality of green innovation (Mijit et al., 2025). Innovation-management research links corporate digitalization to green technology innovation using text-based measures (Shao et al., 2024). AI deployment has also been shown to support process innovation when firms face constraints in knowledge transfer (Standaert and Andries, 2026). Recent evidence further associates AI investment intensity with the temporal stability of green innovation (Hou et al., 2026). Together, these studies establish the relevance of AI and digital technologies for innovation outcomes, but they mostly measure adoption, capability, digitalization, strategic emphasis, or policy exposure. They say less about whether reported AI enters the workflow where operating decisions are made.

Green innovation is implemented through changes in products and production processes (Rennings, 2000). Green product innovation also depends on organizational capabilities that connect sustainability goals with development and manufacturing activities (Dangelico et al., 2017). A technically accurate signal has little value if it reaches no responsible actor. It also loses value when it arrives after the decision or cannot trigger a revision. The bottleneck can lie between information and action even when information generation is adequate.

We describe the underlying information-to-action architecture as AI-enabled execution capability (AEC). It links AI-supported signals to monitoring and coordinated response, then closes the loop through feedback and revision. The archival measure is narrower: it identifies annual-report sentences that place AI-related tools inside concrete operational or governance workflows. We call this measure *disclosed AEC*. It measures reported operational embedding. It does not assess system quality, use intensity, or employee compliance.

We distinguish two execution problems. Coordination complexity creates interdependence: more actors, systems, and interfaces must align before action occurs. Better information can reduce handoff failures, but it cannot settle authority, incentives, or conflict. Real-time adaptation requirement creates a temporal problem because information loses value while operating conditions change. Galbraith frames organization design as an information-processing problem under uncertainty (Galbraith, 1974). Tushman and Nadler likewise emphasize matching information-processing arrangements to task interdependence (Tushman and Nadler, 1978). Task–technology fit adds a functional criterion: technology creates value when its capabilities match what the task requires (Goodhue and Thompson, 1995). These

perspectives imply a specific boundary condition: operational AI embedding should be most useful when the cost of stale information is high.

We test this argument in an unbalanced panel of 19,018 Chinese A-share firm-year observations from 2020 to 2025. Annual-report measures and controls from fiscal years 2019–2024 are matched to later green-patent applications. Sentence-level rules distinguish disclosed AEC from general AI disclosure (GAI), digital transformation (DT), and intelligent manufacturing (IM). The main specifications use firm fixed effects and Poisson pseudo-maximum-likelihood models. The measurement design requires AI language to appear with execution and embedding features, then checks that distinction against adjacent digital measures, non-text task proxies, and external environmental-enforcement pressure.

The results shift attention from average AI disclosure to task fit. The within-firm AEC coefficient attenuates after demanding controls, whereas the interaction with real-time adaptation requirement remains positive across benchmark adjustment and several alternative specifications. After benchmark adjustment, the interaction-family value is $q = 0.074$, meeting the 10% false-discovery-rate threshold, while the corresponding coordination interaction is less precise. The difference between the two interaction coefficients is suggestive ($p = 0.091$). This reframes the technology-management question around the execution conditions that allow AI-generated information to retain value for innovation.

The paper combines a narrower measure with a sharper contingency. It shifts archival AI measurement from broad attention or adoption toward reported workflow embedding and separates coordination complexity from the temporal problem of rapid adaptation. The more stable conditional pattern appears with adaptation requirement. The contribution is a measurement-and-fit account of operational AI embedding in innovation.

2. Theoretical Background and Conceptual Development

2.1. Green Innovation as an Information-to-Action Problem

Green innovation covers product and process changes that reduce environmental burdens (Rennings, 2000). Firm-level evidence shows that environmental innovation responds to regulation, technological opportunity, and firm characteristics (Horbach, 2008). Organizational capabilities also shape proactive environmental strategy (Sharma and Vredenburg, 1998). These explanations clarify why firms search for greener solutions. They do not determine whether environmental information changes an operating decision.

Implementation is distributed across functions. Cleaner-process changes may begin in engineering or maintenance but soon affect schedules, suppliers, environmental control, and managerial authorization. Green products create a different chain, linking R&D with manufacturing, procurement, and compliance. In both cases, information has to cross organizational interfaces without becoming stale. Green innovation includes an execution problem.

2.2. From AI Attention to Operational Embedding

AI-capability research conceptualizes AI as a bundle of firm resources and competencies (Mikalef and Gupta, 2021). Digital-transformation research operates at a wider level, covering process, organizational, and business-model change (Verhoef et al., 2021). Manufacturing-focused AI research examines adoption and implementation in production settings (Kinkel et al., 2022). Innovation-management evidence also shows that AI use varies across stages of the innovation process (Roberts and Candi, 2024), making its organizational placement part of the empirical question. These streams capture different layers of technological capacity and organizational change, but none directly identifies whether AI-generated information is embedded in a recurring execution workflow.

For our question, the unresolved step is execution. An emissions forecast cannot reduce emissions unless it enters a decision routine. A predictive-maintenance signal matters only when it changes inspection, scheduling, or intervention. A supplier-risk alert cannot shape product design if it remains outside procurement and engineering workflows. This creates a measurement problem as well. Annual reports can reflect substantive use, managerial attention, or symbolic positioning. A general AI count cannot be assumed to represent operational use (Li et al., 2025).

2.3. AI-Enabled Execution Capability

AEC denotes the organizational architecture through which AI-supported information is connected to monitoring, coordinated response, feedback, and revision:

$$AI \text{ signal} \rightarrow \text{Monitoring} \rightarrow \text{Coordination} \rightarrow \text{Feedback} \rightarrow \text{Revision}. \quad (1)$$

Monitoring identifies an operating state or exception. Coordination assigns responsibility and links action across tasks or systems. Feedback compares the response with the intended result. Revision changes a parameter, schedule, decision, or routine when the observed result is inadequate. These functions form a sequence; they are not interchangeable labels for the same resource.

The archival measure captures the observable part of this architecture. A sentence is retained when AI-related content appears together with an execution feature and operational or governance embedding. Generic AI aspirations, broad digital-upgrading statements, automation without an AI link, and smart-manufacturing language without a workflow connection are excluded. The resulting variable measures disclosed operational embedding. It does not assess the quality of the underlying system.

This boundary separates AEC from adjacent measures. GAI captures prediction, recognition, classification, recommendation, and analytical support. DT captures broad digitization and process integration. IM captures production automation and smart-factory control. Because a sentence can contain more than one theme, the joint models ask whether AEC adds information after broader digital language is considered.

Corporate filings contain information that can be extracted systematically through textual analysis (Li, 2010). Text-as-data methods can recover structured signals from large document collections, but construct validity depends on how the measure is defined (Gentzkow et al., 2019). Innovation research has also used natural language processing to structure large textual knowledge bases for innovation search (Just, 2024). The empirical analysis here uses text to operationalize disclosed workflow embedding; direct observation of use intensity and response behavior remains outside the data.

2.4. Why Adaptation Requirement Differs from Coordination Complexity

Organizational information-processing theory treats organization design as a response to information-processing demands (Galbraith, 1974). Task interdependence raises those demands and increases the need for appropriate information-processing arrangements (Tushman and Nadler, 1978). Task–technology fit adds a functional criterion: technology creates value when its capabilities match what the task requires (Goodhue and Thompson, 1995). Research on IT business value reaches a similar conclusion: value depends on processes and complementary organizational assets, not technology alone (Melville et al., 2004).

Coordination complexity is mainly an interdependence and governance friction. As actors and interfaces multiply, information can be delayed or detached from responsibility. Embedded AI may improve visibility and continuity, but it cannot determine authority or settle conflicts over incentives. Its fit with coordination complexity is partial.

Real-time adaptation requirement is mainly a temporal friction. Operating states, emissions readings, energy loads, supplier conditions, and regulatory requirements can change while implementation is under way. Information loses value as it ages. Monitoring, anomaly detection, rapid updating, and adaptive scheduling directly address this loss of timeliness. We expect the fit to be stronger when decisions must be revised before the underlying information becomes stale.

2.5. Conceptual Framework

Figure 1 isolates the relationships tested in the empirical analysis. The AEC–green innovation association is the baseline path, while coordination complexity and real-time adaptation requirement enter as distinct moderators; H4 compares the strength of those two contingencies. Green strategic intention remains contextual and is not drawn as an estimated antecedent. GAI, DT, and IM are used later as empirical benchmarks and sit outside the causal framework.

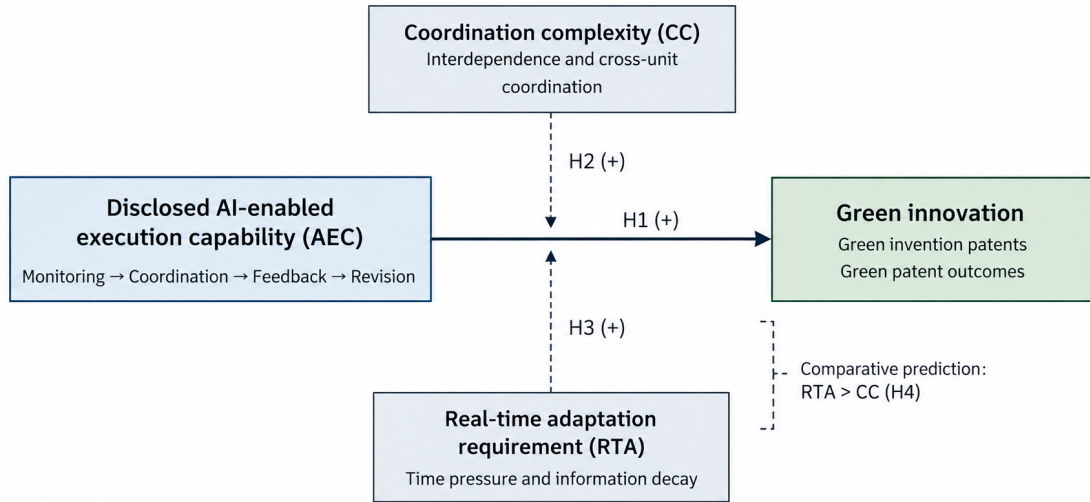


Figure 1. Conceptual framework linking disclosed AI-enabled execution capability to green innovation under coordination complexity and real-time adaptation requirement.

3. Hypothesis Development

3.1. Baseline Association

Operationally embedded AI can connect environmental signals with the routines that govern maintenance, scheduling, resource allocation, and corrective action. Feedback from those routines can reveal recurrent losses or technical constraints. Some of these problems remain operational; others become inputs to experimentation, creating a plausible route from execution data to inventive activity.

Environmental innovation also depends on regulation, technological opportunity, and firm-specific resources (Horbach, 2008). Complementary organizational assets shape whether environmental practices translate into performance (Christmann, 2000). Disclosure further varies in how closely it reflects actual use. We expect a positive average association, with variation across firms and settings.

H1. Disclosed AI-enabled execution capability is positively associated with green innovation.

3.2. Coordination Complexity

Green innovation becomes coordination intensive when technical work must pass between R&D, production, procurement, suppliers, compliance staff, and managers. Every additional interface creates another opportunity for delay or inconsistent interpretation. Workflow-embedded AI can reduce some of these failures by preserving information continuity and making follow-up visible across task stages (Modgil et al., 2022).

Coordination also depends on authority, incentives, conflict resolution, and agreed routines. AI can expose interdependence without resolving the governance dispute that the information reveals. H2 predicts a positive moderation effect. The functional match remains incomplete.

H2. The association between disclosed AEC and green innovation is stronger when coordination complexity is higher.

3.3. Real-Time Adaptation Requirement

Real-time adaptation requirement concerns how quickly information loses value during implementation. Emissions readings, equipment conditions, energy loads, supplier behavior, and regulatory signals can change before a planned response is executed. In these settings, a delayed signal can be technically correct and still be operationally useless.

Embedded AI is closely matched to this problem because it can connect repeated monitoring with scheduling, maintenance, process control, resource reallocation, or corrective action. Its value should rise when operating conditions change frequently and decisions must be revised before the information becomes stale.

H3. The association between disclosed AEC and green innovation is stronger when the real-time adaptation requirement is higher.

The two moderators imply a comparative prediction. Coordination complexity includes governance frictions that better information cannot remove. Adaptation requirement is tied more directly to the speed and continuity of information. The fit with AEC should be stronger for adaptation requirement.

H4. Real-time adaptation requirement strengthens the AEC–green innovation association more than coordination complexity does.

4. Research Design

4.1. Research Setting and Sample

The outcome panel covers non-financial Chinese A-share listed firms from 2020 to 2025. Financial firms are excluded because their regulation, balance sheets, and patenting processes differ from those of industrial and service firms. Special-treatment observations and firm-years without usable annual-report text, reliable identifiers, or complete lagged controls are also excluded. Starting from 24,936 candidate firm-years, text-quality screening and database matching yield an unbalanced panel of 19,018 observations. Appendix Table 10 reports the sample construction.

Patent counts are assigned by filing year. For firm-years successfully matched to the patent database, no green-patent application is coded as zero. An observation is excluded only when the match is unreliable or a required covariate is unavailable. Non-patenting firms remain in the estimation sample.

China offers substantial variation in industrial digitalization, environmental enforcement, industrial structure, and corporate disclosure. It also creates a measurement risk because listed firms may discuss AI and green development for policy-facing or legitimacy reasons. We treat disclosure style as a competing explanation and do not assume that technical language represents operating practice.

Annual-report measures and covariates dated 2019–2024 are matched to green-innovation outcomes dated 2020–2025. The database was frozen on 31 May 2026. The main ordering uses a one-year fiscal lag. Because a fiscal-year label does not guarantee that the report was released before a patent filing, a calendar-ordered sample retains only patent filings after the corresponding report-release date. Two-year lags provide an additional timing check.

4.2. Data Sources

Financial and governance variables come from CSMAR and WIND. Annual reports are collected from CNINFO and matched by stock code and fiscal year. Green patent records come from CNRDS and are classified at the application level using International Patent Classification categories for environmentally relevant technologies. We use applications because filing dates are closer to the timing of inventive effort than grant dates. Name changes and affiliated applicants are reconciled before aggregation.

Environmental-enforcement records are used to construct external adaptation pressure at the province–industry–year level. Business segments, majority-owned subsidiaries, supplier concentration, quarterly sales, and operating costs provide non-text measures of coordination complexity and industry adaptation requirement. These variables are used for source separation, not as causal treatments.

4.3. Annual-Report Processing

Annual-report PDFs are converted to machine-readable Chinese text. Embedded text is extracted directly where available, while image-only pages are processed with optical character recognition. Page headers, page numbers, non-narrative tables, exact duplicate lines, and repeated boilerplate are removed before character normalization and sentence segmentation. Matches are evaluated within the same cleaned sentence. Negated statements and statements that refer only to future intentions are excluded when the qualifier governs the matched clause.

The protocol is a rule-based natural-language processing pipeline built from Chinese dictionaries and regular-expression families, with explicit sentence, negation, future-intention, and document-cleaning rules. Appendix Table 13 gives English glosses of the matching logic. The full Chinese dictionaries and regular-expression rules are available from the corresponding author for reproducibility checks, subject to source-data licensing constraints.

4.4. Variable Measurement

4.4.1. Green innovation

The main dependent variable, $GI1$, is the natural logarithm of one plus green invention patent applications. $GI2$ is the natural logarithm of one plus all green patent applications, including invention and utility-model patents. $GI3$ is a supplementary intensity measure defined as $GreenPatentApplications / (TotalPatentApplications + 1)$. Green invention applications are the primary outcome because they are more likely than utility models to represent substantive technological development. Count models use the untransformed patent outcomes.

4.4.2. Disclosed AI-enabled execution capability

The main explanatory variable is a sentence-level disclosure proxy for AEC. A retained sentence must contain an AI-related term, an execution-oriented feature, and operational or governance embedding. The execution layer covers monitoring, scheduling, task coordination, workflow progression, process optimization, cross-system operation, and feedback-based adjustment. The embedding layer links these features with production, supply chains, energy, emissions, operational control, compliance, or decision routines.

Generic statements about AI opportunity, digital upgrading, intelligent transformation, smart manufacturing, or automation are excluded when no AI-workflow connection is present. Future-oriented statements are excluded unless they describe an existing execution or governance routine. The raw score is the number of valid sentences per thousand report sentences and is standardized for estimation:

$$AEC_{i,t}^{raw} = 1000 \times \frac{N(AI \cap Execution \cap Embedding)_{i,t}}{N(Sentences)_{i,t}}, \quad AEC_{i,t} = \frac{AEC_{i,t}^{raw} - \overline{AEC^{raw}}}{sd(AEC^{raw})}. \quad (2)$$

The same scaling rule is applied to the other text indices.

4.4.3. Benchmark and execution-requirement variables

General AI disclosure (GAI) covers prediction and recognition tasks together with classification, recommendation, and analytical support. Digital transformation (DT) captures broad digitization and process integration, whereas intelligent manufacturing (IM) is closer to production automation and smart-factory control. These benchmarks help separate operational embedding from broader technical attention. Their joint coefficients are interpreted as incremental associations; full orthogonality is not assumed.

Disclosed coordination complexity (CC) is formed from sentences about cross-functional coordination, system integration, supply-chain linkage, and integrated process management. Disclosed real-time adaptation requirement (RTA) is formed from sentences about real-time monitoring, adaptive scheduling, closed-loop response, rapid iteration, and feedback-based operational adjustment. These measures can reflect task requirements, established practice, or managerial attention. We triangulate them with non-text proxies. An overlap-excluded RTA specification also removes terms shared with the AEC coding layers.

The structural CC proxy is the equal-weight mean of standardized log business-segment count, log majority-owned subsidiary count, and inverse top-five supplier concentration. The industry RTA proxy combines trailing three-year volatility in quarterly sales growth and operating-cost growth at the two-digit industry level.

External adaptation pressure (EAP) is constructed at the province–industry–year level from environmental penalty cases, real penalty amounts, and inspection records. Penalty amounts are converted to constant 2019 RMB. Each component is scaled by industrial activity, transformed, differenced, and standardized within year:

$$q_{p,j,t}^k = \log \left(1 + \frac{Enforcement_{p,j,t}^k}{IndustrialScale_{p,j,t}} \right), \quad k \in \{cases, amount, inspection\}, \quad (3)$$

$$EAP_{p,j,t} = \frac{1}{3} \sum_k Std_t \left(q_{p,j,t}^k - q_{p,j,t-1}^k \right). \quad (4)$$

Higher values indicate a larger increase in enforcement intensity. EAP is an external contextual measure, not a treatment.

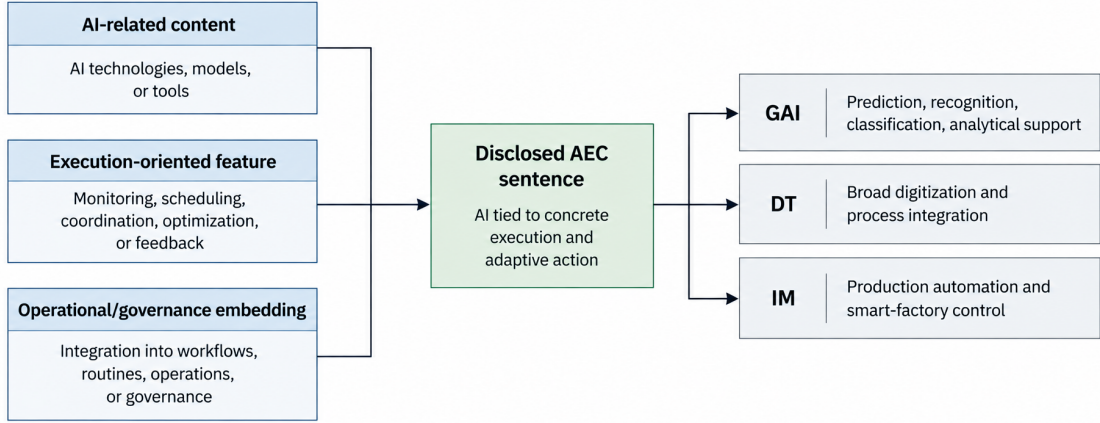


Figure 2. Measurement logic and construct boundaries for disclosed AEC and adjacent digital measures. The links to GAI, DT, and IM mark construct-boundary comparisons; they are not causal paths.

4.4.4. Supplementary outcomes and controls

Environmental performance is the designated secondary outcome because it is closest to the proposed operational mechanism. Green total factor productivity (GTFP), environmental investment, and green patent quality are exploratory outcomes. None is interpreted as a mediator.

Controls include firm size, age, profitability, leverage, R&D intensity, Tobin's Q, state ownership, board independence, industry competition, and provincial environmental-regulation intensity. Continuous firm-level variables are winsorized at the 1st and 99th percentiles by year. Explanatory variables and controls enter with a one-year lag, and the control set is held constant within each model comparison. Appendix Table 12 gives the full variable definitions.

4.5. Construct Validation and Measurement Checks

The main measurement threat is common disclosure style. Firms with longer, more technical, or more environmentally oriented reports may score highly on several text measures even when the underlying practices differ. Construct-specific rules and overlap-excluded RTA reduce lexical reuse. Observable document style is addressed with report length, readability, tone, green-disclosure intensity, and the adjacent digital measures; a residualized AEC measure removes additional report-characteristic and industry-year variation. Source dependence is examined separately with structural CC, industry RTA, and EAP. Future AEC, a randomized pseudo-dictionary, and non-environmental patent outcomes provide falsification checks.

Two coders independently assessed 400 AEC and 360 GAI candidate sentences, together with 320 candidates for each moderator. Across constructs, inter-coder agreement ranges from 0.83 to 0.86 and Cohen's kappa from 0.70 to 0.72. A separate stratified holdout audit yields precision of 0.81–0.88; recall is 0.76–0.85. These checks assess coding consistency and retrieval quality only. They cannot validate internal system use.

4.6. Econometric Models

The baseline fixed-effects model is

$$GI_{i,t} = \beta_1 AEC_{i,t-1} + \gamma' X_{i,t-1} + \mu_i + \lambda_t + \epsilon_{i,t}, \quad (5)$$

where μ_i and λ_t denote firm and year fixed effects. Standard errors are clustered by firm (Petersen, 2009). Because green patent counts are non-negative and contain many zeros, fixed-effects Poisson pseudo-maximum-likelihood models are estimated on raw counts (Santos Silva and Tenreyro, 2006). The high-dimensional fixed-effect implementation follows the computational approach described by Correia et al. (Correia et al., 2020):

$$E(\text{GreenPatentCount}_{i,t} | Z_{i,t-1}) = \exp(\beta_1 AEC_{i,t-1} + \theta' Z_{i,t-1} + \mu_i + \lambda_t). \quad (6)$$

The moderation model is

$$GI_{i,t} = \beta_1 AEC_{i,t-1} + \beta_2 CC_{i,t-1} + \beta_3 RTA_{i,t-1} + \beta_4 (AEC_{i,t-1} \times CC_{i,t-1}) + \beta_5 (AEC_{i,t-1} \times RTA_{i,t-1}) + \gamma' \mathbf{X}_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t}. \quad (7)$$

Because AEC, CC, and RTA are standardized, β_1 is the AEC association at the mean of the two moderators. The marginal effect is

$$\frac{\partial GI_{i,t}}{\partial AEC_{i,t-1}} = \beta_1 + \beta_4 CC_{i,t-1} + \beta_5 RTA_{i,t-1}. \quad (8)$$

The surface reported in the results combines two pairwise interactions; it does not estimate a three-way interaction (Brambor et al., 2006).

The source-separated model replaces report-based adaptation requirement with EAP:

$$GI_{i,t} = \alpha + \beta_1 AEC_{i,t-1} + \beta_2 EAP_{pjt-1} + \beta_3 (AEC_{i,t-1} \times EAP_{pjt-1}) + \gamma' \mathbf{X}_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t}. \quad (9)$$

GII with firm and year fixed effects is the primary outcome. $AEC \times RTA$ is the theory-prioritized interaction, while $AEC \times CC$ provides the mechanism contrast. In the joint AEC–GAI model, the four interaction terms form one Benjamini–Hochberg family (Benjamini and Hochberg, 1995). We report raw p-values, adjusted q-values, confidence intervals, and the Wald test that compares the two AEC interactions.

4.7. Identification and Scope of Inference

The design is observational. Fiscal-year lags, firm fixed effects, count models, source-separated moderators, report-style adjustment, timing restrictions, and negative controls reduce specific competing explanations but cannot remove time-varying confounding. Environmental strategy, digital maturity, managerial attention, or disclosure quality may affect both the text measures and patenting. Calendar-ordered and two-year-lag models improve temporal ordering but do not identify the causal effect of AI deployment. The estimand is a conditional association involving disclosed operational embedding.

The main specifications use firm-clustered standard errors. EAP varies at the province–industry–year level, so EAP models also report two-way clustering by firm and province–industry as a sensitivity check (Cameron et al., 2011).

5. Results

5.1. Descriptive Evidence

Table 1 reports descriptive statistics and selected correlations. The median number of green patents is zero for both main patent outcomes, which makes count models especially relevant. AEC and GAI are positively related but not interchangeable. AEC is also correlated with CC and RTA, so the interaction analysis must address shared disclosure style and source dependence.

Figure 3 shows the annual descriptive series on a common 2020 base. By 2024, the AEC index reaches 176.6 and GAI 162.8, compared with 133.4 for green innovation. The series are descriptive and do not identify a within-firm effect. All three plotted series use pre-standardization annual measures.

5.2. Average Association and Benchmark Adjustment

Table 2 shows that lagged AEC is positive in pooled OLS and PPML. The coefficient falls in the firm fixed-effects model and becomes statistically indistinguishable from zero after prior innovation is included. H1 is not supported as a stable within-firm average relationship, directing attention to the task-contingent specifications.

Table 3 adds GAI, DT, and IM. GAI is positive when entered alone, whereas both AEC and GAI shrink once they enter jointly and broader digital disclosure is controlled. The result shows that the average AEC coefficient is modest once adjacent digital language is considered. Figure 4 summarizes the within-firm linear estimates. Pooled OLS remains in Table 2 as a descriptive benchmark, while PPML remains in the tables because its count-model coefficient is on a different scale from the plotted linear estimates.

Table 1. Descriptive statistics and selected correlations

<i>Panel A. Descriptive statistics</i>							
Variable	N	Mean	SD	P25	Median	P75	Max
GI1	19,018	0.526	0.971	0.000	0.000	0.693	5.493
GI2	19,018	0.831	1.224	0.000	0.000	1.386	6.397
GI3	18,047	0.094	0.187	0.000	0.000	0.108	1.000
AEC	19,018	0.000	1.000	-0.548	-0.208	0.325	4.684
GAI	19,018	0.000	1.000	-0.596	-0.158	0.415	5.171
CC	19,018	0.000	1.000	-0.493	-0.149	0.358	4.315
RTA	19,018	0.000	1.000	-0.516	-0.198	0.302	4.781
<i>Panel B. Selected pairwise correlations</i>							
Variable	GI1	AEC	GAI	CC	RTA		
GI1	1.000						
AEC	0.129	1.000					
GAI	0.167	0.508	1.000				
CC	0.107	0.368	0.251	1.000			
RTA	0.143	0.344	0.276	0.409	1.000		

Notes. Outcomes cover 2020–2025. AEC, GAI, CC, and RTA are standardized text measures; correlations are unadjusted.

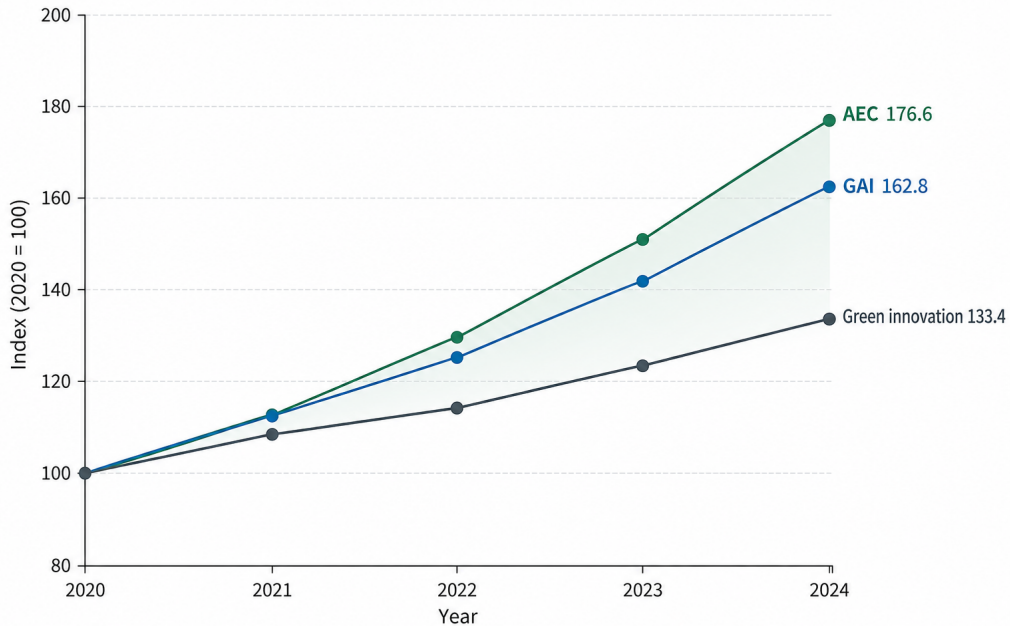


Figure 3. Annual evolution of disclosed AEC, GAI, and green innovation, indexed to 2020 = 100.

5.3. Execution-Requirement Interactions

Table 4 reports the conditional tests. $AEC \times RTA$ is positive in the separate, joint, benchmark-adjusted, and PPML specifications. In Column (4), the coefficient is 0.066 with a standard error of 0.028. Its Benjamini–Hochberg adjusted value is $q = 0.074$. H3 receives qualified support at a 10% false-discovery-rate threshold, but not at 5%.

$AEC \times CC$ is also positive. After benchmark adjustment, the coefficient is 0.051 with a standard error of 0.029 and $q = 0.157$, providing weaker evidence for H2. A Wald test comparing the two AEC interactions gives $\chi^2 = 2.86$ and $p = 0.091$. The point estimates favor adaptation requirement. The formal comparison does not reach conventional 5% significance.

Table 2. Baseline estimates

Variables	(1) OLS + Year FE	(2) Firm FE	(3) Firm FE + lag DV	(4) PPML FE
AEC(t-1)	0.069*** (0.022)	0.039* (0.022)	0.026 (0.020)	0.054** (0.022)
GII(t-1)	— —	— —	0.311*** (0.035)	— —
Controls	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	19,018	19,018	15,748	19,018
R ² / pseudo R ²	0.145	0.076	0.171	0.121

Notes. Firm-clustered standard errors are in parentheses. PPML uses raw patent counts; linear models use GII. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3. Benchmark adjustment with adjacent digital measures

Variables	(1) GAI	(2) AEC	(3) AEC + GAI	(4) Benchmarks	(5) PPML
GAI(t-1)	0.052** (0.022)	— —	0.038* (0.022)	0.031 (0.024)	0.044* (0.025)
AEC(t-1)	— —	0.039* (0.022)	0.027 (0.021)	0.024 (0.018)	0.041* (0.023)
Digital transformation benchmark	No	No	No	Yes	Yes
Intelligent manufacturing benchmark	No	No	No	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	19,018	19,018	19,018	19,018	19,018
R ² / pseudo R ²	0.073	0.076	0.081	0.087	0.129

Notes. Firm-clustered standard errors are in parentheses. Column (5) is PPML; the other columns use GII. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

With firm fixed effects and benchmark adjustment, a one-standard-deviation increase in RTA changes the estimated marginal AEC association by 0.066 log points; the corresponding change for CC is 0.051. Figure 5 places these fixed-effects estimates beside the separate and joint specifications and also shows the smaller GAI benchmark interactions (0.018 for CC and 0.024 for RTA). The PPML estimates are reported in Table 4 and use raw patent counts. In that model, the AEC $\times$ RTA coefficient of 0.083 corresponds to about an 8.7% change in the conditional expected count for a one-standard-deviation increase in RTA, holding the remaining terms constant.

Figure 6 evaluates Equation 8 using Table 4, Column (3), and displays the resulting association as a continuous surface. The quartile reference values are (−0.493, −0.149, 0.358) for CC and (−0.516, −0.198, 0.302) for RTA. At the low-low reference point, the estimated marginal AEC association is −0.038; it is 0.004 at the medians and 0.067 at the high-high point. The zero contour marks where the fitted association changes sign. This surface simply adds the two pairwise interactions in Equation 8; it is not a three-way interaction.

5.4. Threat-Targeted Robustness and Falsification

Table 5 organizes the main checks by the problem they address. Removing overlapping RTA language leaves the focal interaction close to the benchmark estimate. The non-text industry-volatility proxy is smaller and less precise but points in the same direction. Calendar ordering preserves the positive pattern, as does a two-year lag. External adaptation pressure, measured outside the annual report, also yields a positive AEC interaction.

Table 6 broadens the specification checks. Alternative outcomes and AEC definitions test measurement dependence. Province-year and industry-year fixed effects absorb time-varying geographic and industry shocks. Residualized AEC and overlap-excluded RTA reduce report-style and lexical-overlap concerns. Structural CC and industry RTA provide partial source separation.

The falsification tests clarify the scope of the result. Future AEC and the randomized pseudo-dictionary are small and imprecise, and the interaction for non-environmental utility patents is weak. The AEC $\times$ RTA coefficient for

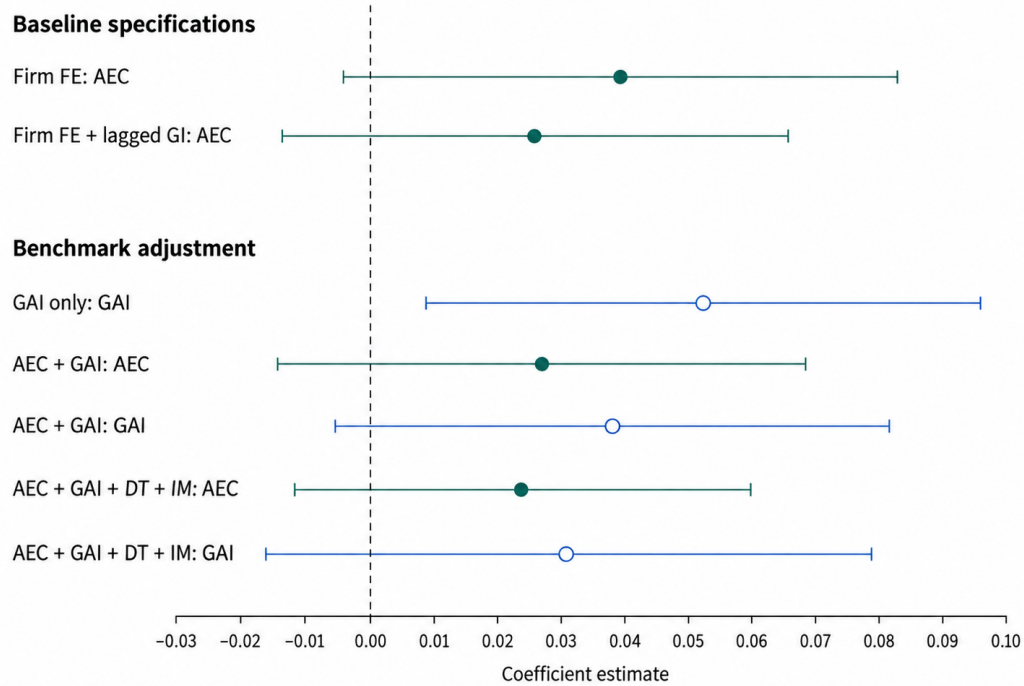


Figure 4. Within-firm AEC and GAI estimates across baseline and benchmark-adjusted specifications. Error bars indicate 95% confidence intervals.

non-green invention patents is positive at the 10% level. The positive non-green estimate leaves open a broader technical problem-solving channel, with green invention as one application.

5.5. Source-Separated Adaptation Evidence

Table 8 replaces report-based adaptation requirement with external adaptation pressure. $AEC \times EAP$ is 0.046 in the fixed-effects model and 0.053 in PPML. With two-way clustering by firm and province–industry, the p-values are 0.038 and 0.041. The result provides source-separated evidence that the adaptation pattern is not confined to annual-report language.

Figure 7 places the source-separated EAP estimates beside the supplementary outcome evidence. The $AEC \times RTA$ coefficients are 0.047 for environmental performance, 0.041 for GTFP, 0.026 for environmental investment, and 0.055 for green patent quality. All point in the same direction. The outcomes and estimators differ, so the coefficient magnitudes are not compared; the common direction is the relevant result.

5.6. Exploratory Heterogeneity

Table 9 and Figure 8 report exploratory subgroup estimates. The full-sample interaction, 0.071, provides a common visual reference. Point estimates are larger in high-pollution industries and manufacturing firms, as well as in settings with stronger environmental regulation and better digital infrastructure; the ownership estimates are almost identical. The subgroup patterns are descriptive because no formal cross-group equality tests are imposed.

Table 4. Coordination and adaptation interactions

Variables	(1) FE + CC	(2) FE + RTA	(3) FE full	(4) FE full + GAI	(5) PPML full
AEC(t-1)	0.034 (0.021)	0.031 (0.020)	0.026 (0.018)	0.021 (0.018)	0.045* (0.024)
CC(t-1)	0.051** (0.024)	–	0.040* (0.022)	0.037* (0.022)	0.036 (0.025)
AEC × CC	0.064** (0.027)	–	0.055* (0.031)	0.051* (0.029)	0.047* (0.027)
RTA(t-1)	–	0.060*** (0.021)	0.048** (0.022)	0.045** (0.021)	0.069*** (0.024)
AEC × RTA	–	0.079*** (0.025)	0.071** (0.030)	0.066** (0.028)	0.083*** (0.026)
GAI(t-1)	–	–	–	0.028 (0.021)	0.039* (0.023)
GAI × CC	–	–	–	0.018 (0.015)	0.013 (0.016)
GAI × RTA	–	–	–	0.024 (0.018)	0.021 (0.019)
Controls / FE	Yes	Yes	Yes	Yes	Yes
N	19,018	19,018	19,018	19,018	19,018
R ² / pseudo R ²	0.089	0.094	0.101	0.106	0.135

Notes. Firm-clustered standard errors are in parentheses. In Column (4), BH-adjusted q-values are 0.157, 0.074, 0.230, and 0.230 for the four interactions. The AEC interaction equality test gives $p = 0.091$.

Table 5. Threat-targeted checks

Specification	Coefficient	SE	N	Primary threat addressed
AEC × RTA, full FE + GAI	0.066	0.028	19,018	Reference text model
Overlap-excluded RTA	0.061	0.028	19,018	Lexical overlap
Non-text industry RTA proxy	0.046	0.026	19,018	Common report source
Calendar-ordered filing sample	0.063	0.030	16,942	Report-publication timing
Two-year lag	0.058	0.029	15,406	Innovation gestation
AEC × EAP, firm FE	0.046	0.020	19,018	Common text source

Notes. RTA = real-time adaptation requirement; EAP = external adaptation pressure. The reference interaction has BH-adjusted $q = 0.074$.

Table 6. Robustness checks

Specification	AEC main	AEC × CC	AEC × RTA	N
Main DV = GI1	0.026 (0.018)	0.055* (0.031)	0.071** (0.030)	19,018
DV = GI2	0.044** (0.020)	0.067** (0.031)	0.083*** (0.030)	19,018
DV = GI3	0.006 (0.008)	0.017* (0.010)	0.024** (0.011)	18,047
AEC = TF-IDF weighted	0.031* (0.017)	0.049* (0.028)	0.064** (0.027)	19,018
AEC = strict hand-coded	0.018 (0.016)	0.039* (0.023)	0.058** (0.025)	19,018
AEC residualized on report style	0.021 (0.017)	0.045* (0.025)	0.062** (0.028)	19,018
CC = non-text structural proxy	0.024 (0.018)	0.033* (0.020)	–	19,018
RTA = non-text industry proxy	0.025 (0.018)	–	0.046* (0.026)	19,018
Province-year FE	0.019 (0.017)	0.041* (0.024)	0.060** (0.028)	19,018
Industry-year FE	0.021 (0.017)	0.043* (0.025)	0.062** (0.027)	19,018
Excluding 2020	0.023 (0.018)	0.050* (0.028)	0.068** (0.030)	15,863
Ending sample in 2024	0.022 (0.018)	0.047* (0.027)	0.064** (0.029)	15,722
Calendar-ordered filing subsample	0.020 (0.018)	0.046* (0.027)	0.063** (0.030)	16,942
Two-year lag	0.017 (0.019)	0.041 (0.028)	0.058** (0.029)	15,406
RTA = overlap-excluded	0.024 (0.018)	–	0.061** (0.028)	19,018
Report-length and green-disclosure controls	0.020 (0.017)	0.044* (0.025)	0.064** (0.028)	19,018

Notes. Firm-clustered standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

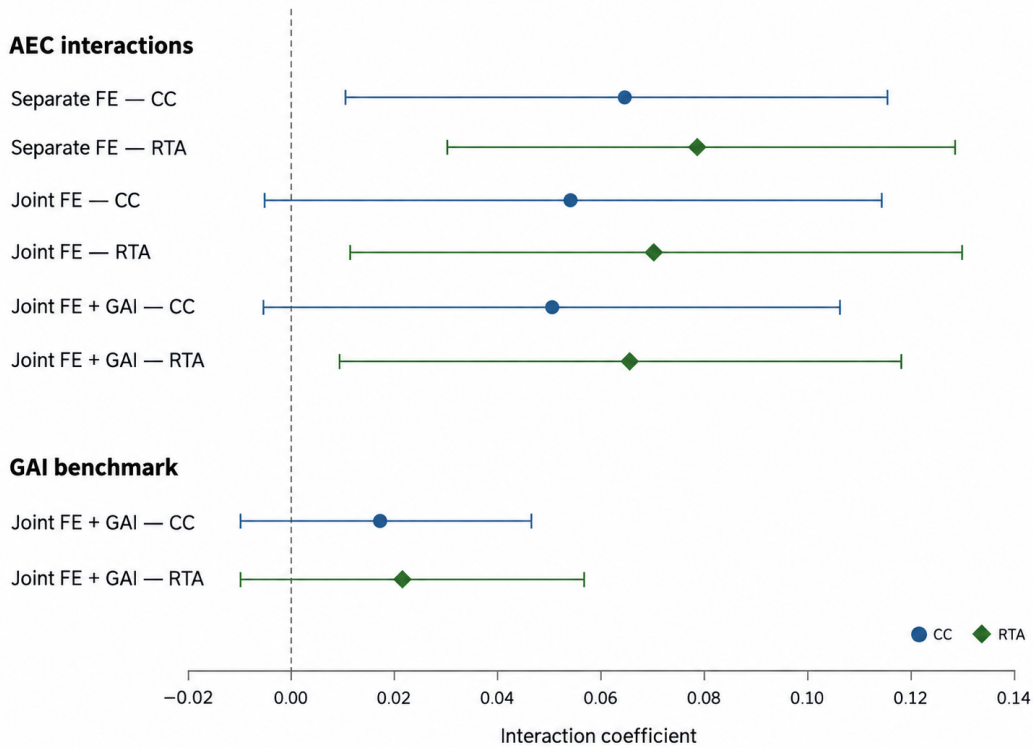


Figure 5. Interaction estimates for coordination complexity and real-time adaptation requirement. Points show fixed-effects coefficients and error bars indicate 95% confidence intervals.

Table 7. Falsification and disclosure-style checks

Outcome / placebo test	AEC main	AEC × CC	AEC × RTA	N
Non-green invention patents	0.018* (0.010)	0.012 (0.018)	0.027* (0.016)	19,018
Future AEC predicting past GI	0.004 (0.014)	0.009 (0.021)	0.011 (0.023)	15,748
Randomized pseudo-AEC dictionary	0.006 (0.013)	0.008 (0.020)	0.010 (0.021)	19,018
Document-length control	0.022 (0.017)	0.046* (0.025)	0.066** (0.028)	19,018
Document length + green disclosure intensity	0.020 (0.017)	0.044* (0.025)	0.064** (0.028)	19,018
Placebo outcome: non-environmental utility patents	0.011 (0.012)	0.006 (0.019)	0.018 (0.017)	19,018
Overlap-excluded RTA measure	0.024 (0.018)	0.049* (0.027)	0.061** (0.028)	19,018
Ending sample in 2024	0.022 (0.018)	0.047* (0.027)	0.064** (0.029)	15,722

Notes. Firm-clustered standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8. External adaptation pressure

Variables	(1) GI1 firm FE	(2) Green patent count PPML
AEC(t-1)	0.024 (0.018)	0.043* (0.023)
EAP(t-1)	0.031* (0.018)	0.047** (0.020)
AEC × EAP	0.046** (0.020)	0.053** (0.023)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	19,018	19,018
R ² / pseudo R ²	0.090	0.130

Notes. Firm-clustered standard errors are in parentheses. Two-way clustering gives $p = 0.038$ and $p = 0.041$ for AEC × EAP.

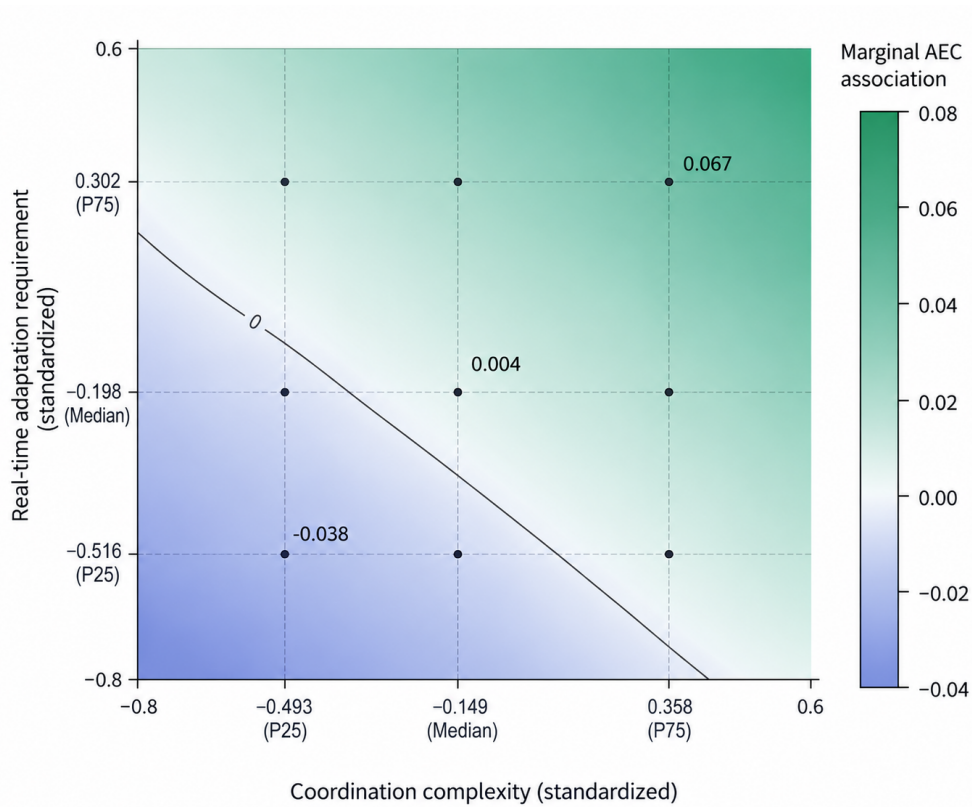


Figure 6. Estimated marginal AEC association across coordination complexity and real-time adaptation requirement. The surface combines the two pairwise AEC interactions from the joint fixed-effects model.

Table 9. Exploratory subgroup estimates

Subgroup	Coefficient	95% CI	N
High-pollution industries	0.083	[0.028, 0.138]	6,657
Other industries	0.047	[-0.006, 0.100]	7,524
Manufacturing	0.077	[0.018, 0.136]	6,292
Non-manufacturing	0.039	[-0.008, 0.086]	8,335
High environmental regulation	0.091	[0.030, 0.152]	9,148
Low environmental regulation	0.043	[-0.014, 0.100]	7,087
High digital infrastructure	0.084	[0.031, 0.137]	10,202
Low digital infrastructure	0.040	[-0.015, 0.095]	8,282
SOEs	0.067	[0.012, 0.122]	6,088
Non-SOEs	0.067	[0.014, 0.120]	5,837
High R&D intensity	0.081	[0.022, 0.140]	7,209
Low R&D intensity	0.043	[-0.012, 0.098]	9,362

Notes. N is the available estimation sample for each subgroup-specific regression and therefore varies across splits. The estimates are exploratory; differences in significance are not formal tests of coefficient differences.

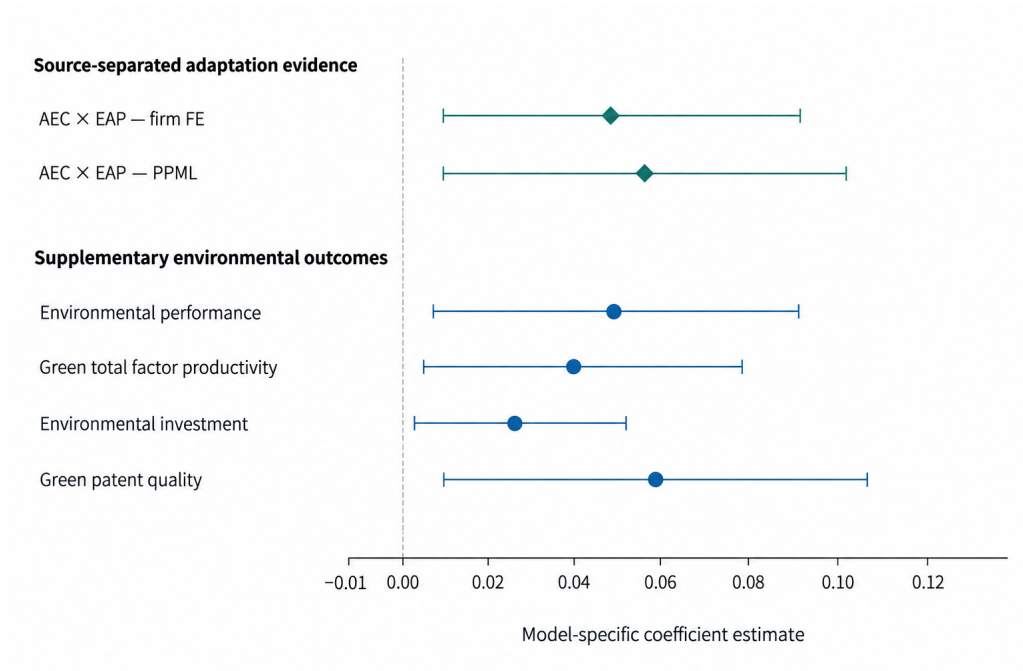


Figure 7. Source-separated adaptation evidence and supplementary environmental outcomes. Points show model-specific coefficient estimates with 95% confidence intervals; magnitudes are not directly comparable across outcomes or estimators.

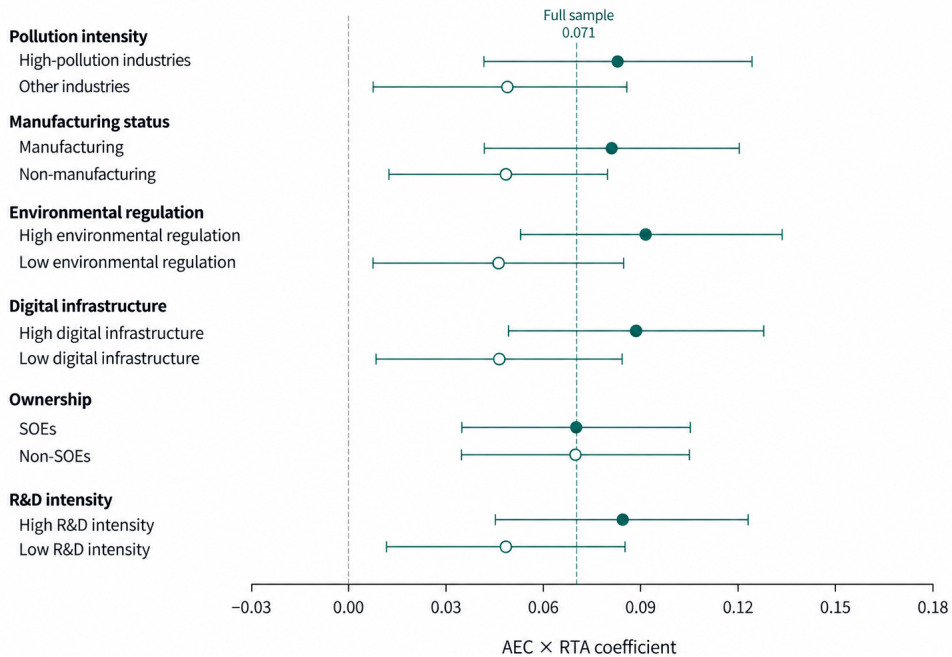


Figure 8. Exploratory subgroup estimates of the AEC × RTA interaction. Error bars show 95% confidence intervals, and the dotted reference line marks the full-sample estimate. No formal cross-group equality test is represented.

6. Discussion

6.1. What the Results Change

The main result lies in the conditional estimates. The average AEC relationship weakens once firm fixed effects, prior innovation, and adjacent digital measures enter the model. The AEC association is stronger when operating conditions require timely adaptation. The empirical focus shifts from AI attention to the execution setting in which AI-generated information can retain value.

Recent annual-report studies measure AI applications with keyword-based or text-derived indices (Chen et al., 2025). Other work treats annual-report AI language as corporate AI focus and does not equate it with verified implementation (Li et al., 2025). Text-based digitalization measures have also been used to examine green technology innovation (Shao et al., 2024). Requiring a workflow connection isolates a more operational layer of disclosure. Its empirical behavior differs from general AI language once task conditions are introduced, making execution embedding a useful dimension for archival technology-management research.

6.2. Temporal Friction Is the More Credible Channel

The distinction between governance friction and temporal friction explains why the two moderators need not behave alike. Coordination complexity can expose disputes over authority and responsibility as well as failures of cross-unit alignment. Better information may reduce handoff failures, yet it cannot settle those disputes. Real-time adaptation requirement poses a narrower problem because information expires while the organization is acting. Continuous monitoring and rapid updating address that problem more directly.

The RTA interaction is larger and more stable than the CC interaction. The equality test gives $p = 0.091$, leaving the formal comparison unresolved. In this sample, temporal adaptation receives stronger empirical support than coordination.

6.3. Operational Learning Beyond Green Innovation

A connection from operating routines to invention requires learning. AI-supported monitoring can expose recurrent anomalies, resource losses, bottlenecks, and environmental constraints. When these observations enter maintenance, production, compliance, or engineering routines, they can redefine technical problems and motivate experiments. That is the plausible route from execution feedback to R&D.

The non-green invention result broadens the interpretation. AEC under high RTA may support general technical problem solving, with green invention as one important application. Future work can compare patent classes directly and trace whether operational feedback is attached to environmental projects or to more general engineering problems.

6.4. Competing Explanations and Boundary Conditions

Digital maturity, managerial technical attention, disclosure quality, or an unobserved environmental strategy could generate both AEC language and later patenting. Report-style controls, adjacent digital measures, and source-separated moderators reduce the most direct disclosure-based alternatives. Time-varying omitted variables remain a limitation of the observational design.

Text-based CC and RTA can describe task requirements, established practice, or managerial attention. Structural CC, industry volatility, and EAP provide complementary source-separated measures with related but distinct content. Their convergence strengthens the task-fit interpretation. More direct process-level measurement is still needed.

The Chinese listed-firm setting further bounds the claim. Policy exposure, disclosure incentives, ownership structures, and digital infrastructure differ from those facing private firms or firms in other institutional systems. The portable claim is the execution-fit logic, not the reported coefficient size.

6.5. Managerial and Policy Implications

For managers, a practical question is where information loses value before action occurs, even when AI tools are already in place. A monitoring system has little operational value if no unit owns the response. It also fails when approval takes longer than the information remains useful or when feedback cannot change the routine. The estimates point most clearly to time-sensitive operating decisions in maintenance and scheduling, process control, resource allocation, supplier follow-up, and compliance.

For investors and regulators, annual-report AI language should not be treated as evidence of operational maturity. More informative disclosure would identify the process in which AI is used, the function responsible for responding, the action triggered by the output, and the feedback that changes later decisions. These details are harder to mimic with generic technology language and more useful for assessing implementation.

The EAP result points to a related policy trade-off. Stronger environmental pressure may raise the value of timely organizational response, but digital infrastructure alone cannot create that response. Responsibility, authority, and process design remain complementary conditions.

6.6. Limitations and Research Priorities

The first limitation is measurement. Disclosed AEC does not reveal system quality, use intensity, response latency, or employee compliance. Procurement records, software-use logs, AI-related vacancies, surveys, and field evidence would provide a stronger test of actual execution.

Second, the theoretical architecture contains monitoring, coordination, feedback, and revision, while the sentence-level measure only requires an AI link to execution-oriented workflow elements. A firm can score highly without showing that all stages operate as one closed loop. Stage-specific measures or process traces would test the construct more strictly.

Third, AEC, CC, and RTA partly share a textual source. Lexical exclusion, document controls, non-text proxies, and EAP reduce but do not eliminate common-source concerns. Plant-, process-, or project-level measures of task requirements would provide a cleaner test.

Fourth, fixed effects and lags do not identify causal deployment effects. Time-varying confounding remains possible, and patent gestation differs across inventions. Implementation dates, procurement shocks, or longer pre- and post-adoption records would provide a stronger design.

Finally, the non-green invention result points to a broader technical problem-solving mechanism. A uniquely green capability interpretation would require formal cross-category tests and project-level links between operational feedback and environmental inventions.

7. Conclusion

Operational AI embedding is not associated with a stable average increase in green invention within firms. Its more consistent association appears when information must be updated and acted on quickly. The conditional association is strongest when monitoring and updating address temporal information decay. Task complexity alone does not show the same pattern.

The study contributes a measurement distinction and a boundary condition. AI attention is not equivalent to operational execution, and execution value depends on the problem the technology is asked to solve. In this sample, the most consistent pattern appears where operating information can lose value quickly, distinguishing temporal adaptation from coordination complexity. The disclosure-based design also points to a broader technical problem-solving process that future work can observe more directly.

A. Measurement and Sample Details

Table 10. Sample construction

Step	Firm-year observations	Treatment
Non-financial A-share candidate firm-years, 2020–2025	24,936	Initial listed-firm outcome panel
After excluding special-treatment and abnormal-trading observations	23,652	Removes observations with exceptional listing status
After annual-report text extraction and quality screening	22,573	Removes missing, image-defective, or non-parsable reports
After identifier, patent, control, industry, and geographic matching	19,018	Final estimation panel; matched firm-years without applications are coded as zero

Notes. The final panel is unbalanced. The displayed sequence follows the dataset-construction order.

Table 11. AEC coding boundary and exclusion rules

Sentence type	Coding decision	Representative sentence pattern	Rationale
Generic AI aspiration	Not AEC	The company explores artificial intelligence opportunities.	No link to an execution process.
Broad digital transformation	Not AEC	The company promotes digital transformation and informatization.	Digital upgrading without an AI-workflow link.
Smart manufacturing without AI linkage	Not AEC	The company advances smart-factory construction.	No explicit AI-enabled monitoring, coordination, or adjustment.
Automation without AI	Not AEC	Automated production lines improve production efficiency.	Automation without AI-related execution content.
Operational AI monitoring	AEC	AI-based monitoring supports equipment maintenance scheduling.	AI is linked to operational monitoring.
AI-based scheduling or optimization	AEC	Intelligent algorithms optimize production scheduling and resource allocation.	AI is linked to scheduling, allocation, or process control.
AI-enabled environmental execution	AEC	AI-based energy monitoring supports adjustment of operating parameters.	AI output is linked to environmental execution.
Governance or feedback embedding	AEC	AI risk-warning outputs are integrated into follow-up correction routines.	AI signals are linked to responsibility, corrective action, or feedback.

Notes. AEC requires AI-related content, execution-oriented activity, and operational or governance embedding in the same cleaned sentence.

Table 12. Variable definitions

Variable	Definition / construction
GI1	Natural logarithm of one plus green invention patent applications.
GI2	Natural logarithm of one plus total green patent applications.
GI3	Green patent applications divided by <i>Total Patent Applications</i> + 1.
AEC	Standardized disclosure proxy; a valid sentence contains AI content, an execution feature, and operational or governance embedding.
GAI	Standardized sentence-scaled index of prediction, classification, recommendation, recognition, or analytical AI use.
CC	Standardized disclosure proxy covering cross-functional coordination, system integration, supply-chain linkage, and integrated process management.
RTA	Standardized disclosure proxy for real-time adaptation requirement, covering real-time monitoring, adaptive scheduling, closed-loop response, rapid iteration, and feedback-based adjustment.
Structural CC proxy	Equal-weight mean of standardized log business-segment count, log majority-owned subsidiary count, and inverse top-five supplier concentration.
Industry RTA proxy	Equal-weight mean of standardized trailing three-year industry volatility in quarterly sales growth and operating-cost growth.
EAP	Equal-weight mean of within-year standardized changes in log size-scaled penalty cases, real penalty amounts, and inspection records at the province–industry–year level.
Environmental performance	Standardized firm-level environmental-performance outcome.
GTFP	Firm-level green total factor productivity measure.
Environmental investment	Firm-level environmental-investment outcome.
Green patent quality	Quality-weighted green patent outcome.
Firm size	Natural logarithm of total assets.
Firm age	Natural logarithm of one plus years since establishment.
ROA	Net income divided by average total assets.
Leverage	Total liabilities divided by total assets.
R&D intensity	R&D expenditure divided by lagged operating revenue.
Tobin's Q	Market value of equity plus book liabilities divided by total assets.
State ownership	Indicator for ultimate state control.
Board independence	Independent directors divided by total board size.
Industry competition	One minus the revenue-based Herfindahl index within the two-digit industry-year.
Environmental regulation	Provincial pollution-control investment divided by secondary-industry value added, standardized within year.

Table 13. Dictionary logic and English glosses

Construct	Dictionary layer / category	English glosses and matching logic
AEC	AI base terms	artificial intelligence; machine learning; intelligent algorithm; intelligent system; industrial intelligence; operational AI
AEC	Execution-oriented features	monitoring; scheduling; task coordination; workflow orchestration; process optimization; cross-system operation; operational execution
AEC	Operational or governance embedding	production process; supply-chain system; energy management; emissions monitoring; operational control; decision workflow; feedback routine
GAI	General analytical AI terms	prediction; classification; recommendation; image recognition; pattern recognition; decision support; data mining
CC	Coordination-related terms	cross-functional coordination; system integration; supplier collaboration; process linkage; multi-department alignment
RTA	Adaptation-related terms	real-time monitoring; adaptive scheduling; closed-loop response; rapid iteration; feedback-based operational adjustment
Matching rule	Same-sentence conjunction	AEC requires at least one term from all three layers in the same cleaned sentence; CC, RTA, and GAI use separate dictionaries.
Exclusion rule	Negation and prospective language	Negated and purely future-oriented statements are excluded when the qualifier applies to the matched clause.

Notes. Entries are English glosses of the Chinese coding logic; the same rules are applied across firms and years.

Declarations

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. Data are available from the authors upon reasonable request.

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Operational AI Embedding and Green Innovation: When Rapid Adaptation Matters

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data are available from the authors upon reasonable request.

Dear Guest Editors,

Please consider our Research Article, “*Operational AI Embedding and Green Innovation: When Rapid Adaptation Matters*,” for the Technovation Special Issue “*Text as Knowledge for Innovation Management: Ensuring Research Relevance and Rigor with NLP and LLMs*.”

The paper addresses a measurement and technology-management problem in research on artificial intelligence in firms. Broad AI attention in corporate reports does not show whether AI has entered the routines through which information is monitored, acted on, and revised. We develop disclosed AI-enabled execution capability (AEC), a sentence-level construct that retains a passage only when AI content is tied to an execution feature and to an operational or governance workflow. This separates reported operational embedding from general AI disclosure, digital transformation, and intelligent manufacturing.

Using 19,018 Chinese A-share firm-year observations from 2020 to 2025, we find little evidence of a stable average within-firm AEC association after demanding controls. The clearer pattern is conditional: AEC is more strongly associated with subsequent green invention when operating conditions require rapid adaptation. The result remains directionally consistent under timing restrictions, non-text task proxies, and an external measure of environmental-enforcement pressure. The paper therefore contributes both a construct boundary for archival AI research and a task-fit explanation for when operational AI embedding is most relevant to innovation.

Technovation has published related work on natural language processing for innovation search (Just, 2024), corporate digitalization and green technology innovation (Shao et al., 2024), AI use across innovation stages (Roberts and Candi, 2024), and AI-enabled process innovation (Standaert and Andries, 2026). Our study extends this work by using corporate text to identify a specific organizational construct and testing its innovation consequences under distinct execution conditions. The Special Issue is relevant to this design because it invites research that uses textual data to measure theoretical constructs in innovation management.

The manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests.

Thank you for considering the manuscript.

Sincerely,

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Highlights

- Annual-report text distinguishes AI attention from operational embedding.
- The average within-firm AEC association weakens after demanding controls.
- AEC is more strongly linked to green invention under rapid adaptation.
- Non-text and external checks support the adaptation-based pattern.

Declaration of interests

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☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: